



LESSON 5-5

Function Operations

Lesson Overview

Objective

Students will be able to:

- ✓ Combine functions by addition, subtraction, multiplication, or division and identify the domain of the result.
- ✓ Compose functions, specifying the order in which the functions are applied and describing the domain of the composite function.

Essential Understanding

Functions can be combined by operations (+, −, ×, ÷) and by composition. The result of the operation or composition can be described as a single function. The domain of the result may be different from the domains of the original functions.

Previously in this course, students:

- Evaluated individual functions for numerical inputs.
- Described the domains (and ranges) of individual functions.

In this lesson, students:

- Combine functions by operations (+, −, ×, ÷) and compositions and describe the domain of the resulting function.

Increasing Coherence Examples 4, 5, and 6 are not necessarily expected to be addressed on high-stakes assessments.

Later in this topic, students will:

- Analyze composed functions, including identities and inverses of compositions of functions.

This lesson emphasizes a blend of *conceptual understanding* and *application*.

- Students develop and use notation for applying operations to combine functions and to compose functions. They also explain when and why the domain of a combined or composite function is not the same as the domains of the original functions.
- Students apply combinations and composites of functions to model situations such as discounts and coupons.



Glossary is available at SavvasRealize.com.

Vocabulary Builder

REVIEW VOCABULARY

English | Spanish

- **domain** | *dominio*
- **substitution** | *sustitución*

NEW VOCABULARY

- **composite function** | *funcion compuesta*
- **composition of functions** | *composición de funciones*

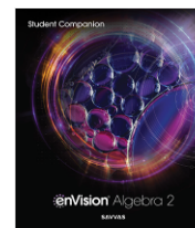
VOCABULARY ACTIVITY

Have students review aspects of substitution to prepare them for learning the composition of functions. To start, ask them to discuss how they can use substitution to check a solution to an equation. **[Replace the variable and see if the result is a true statement.]** Then help students extend that idea of substitution by showing them the function $f(x) = (x + 1)^2$ and asking them to explain how to replace x with each of the following.

1. $3a$ $[f(3a) = (3a + 1)^2 = 9a^2 + 6a + 1]$
2. $r + s$ $[f(r + s) = (r + s + 1)^2 = r^2 + s^2 + 2rs + 2r + 2s + 1]$

Student Companion

Students can do their work for the lesson in their *Student Companion* or on Savvas Realize.



Mathematics Overview

Students add, subtract, multiply, divide, and compose functions. With function composition, they also determine the correct order in which to perform the functions. Then students identify the domain of the resulting functions.

Students use verbal descriptions and algebraic representations to describe the addition, subtraction, multiplication, division, and composition of functions.

Applying Math Practices

Attend to Precision

Students identify and focus on the individual terms of pairs of functions. They do this when combining like terms and when multiplying each term of one function by each term of another function. Students apply this same focus when simplifying quotients and compositions of functions.

Look For and Make Use of Structure

Students use properties of operations on polynomials to combine and compose functions.



MODEL & DISCUSS

INSTRUCTIONAL FOCUS Students explore what it means to combine two functions. They see that the combination (by subtraction) has a practical meaning, and they express the combination of two functions as a single function.

STUDENT COMPANION Students can complete the *Model & Discuss* activity in their *Student Companion*.

Before **WHOLE CLASS**

Implement Tasks that Promote Reasoning and Problem Solving **ETP**

Q: What do the terms *revenue* and *cost* mean? [Revenue is income; cost is an expense of the company.]

Q: How is profit related to revenue and cost? [Profit equals revenue minus cost.]

During **SMALL GROUP**

Support Productive Struggle in Learning Mathematics **ETP**

Q: What is the difference between a fixed cost and a variable cost?

[A fixed cost does not change, while a variable cost changes based on the number of items involved.]

Q: Do both the revenue and the cost functions have both fixed and variable terms?

[No, the cost function has a fixed term and a variable term, but the revenue function has only a variable term.]

For Early Finishers

Q: Last month, Grooming USA made a profit of \$300. What will their profit be if they double the number of pets they groom this month? How many pets do they have to groom this month to double last month's profit? [\$1,600; 160 pets]

Q: Does their profit double when the number of pets that are groomed doubles? [No; they only need to groom 30 more pets to double their profit. When they double the number of pets they groom, their profit is increased by more than 5 times.]

After **WHOLE CLASS**

Facilitate Meaningful Mathematical Discourse **ETP**

Q: How many functions were generated in this activity [two functions: one for revenue and one for cost]

Q: How can the two functions be combined to make a profit function? [by subtracting the function for cost from the function for revenue]

HABITS OF MIND

Use with MODEL & DISCUSS

Make Sense and Persevere A business "breaks even" when its revenue equals its costs. How many pets would Grooming USA have to groom in order to break even?

[100 pets]



Critique & Explain is available at SavvasRealize.com.

Go back

eMODEL & DISCUSS

In business, the term *profit* is used to describe the difference between the money the business earns (revenue) and the money the business spends (cost).

A. Grooming USA charges \$25 for every pet that is groomed. Let x represent the number of pets groomed in a month. Define a revenue function for the business.

Enter your answer:

B. Materials and labor for each pet groomed cost \$15. The business also has fixed costs of \$1,000 each month. Define a cost function for this business.

Enter your answer:

C. Last month, Grooming USA groomed 95 pets. Did they earn a profit? What would the profit be if the business groomed 110 pets in a month?

Enter your answer:

STUDENT EDITION

5-5 Function Operations

I CAN... perform operations on functions to answer real-world questions.

VOCABULARY

- composite function
- composition of functions

MODEL & DISCUSS

In business, the term *profit* is used to describe the difference between the money the business earns (revenue) and the money the business spends (cost).

A. Let x represent the number of pets groomed in a month at Grooming USA. Define a revenue function for the business.

B. Materials and labor for each pet groomed cost \$15. The business also has fixed costs of \$1,000 each month. Define a cost function for this business.

C. Last month, Grooming USA groomed 95 pets. Did they earn a profit? What would the profit be if the business groomed 110 pets in a month?

D. **Generalize** Explain your procedure for calculating the profit for Grooming USA. Suppose you wanted to calculate the profit for several different scenarios. How could you simplify your process?

ESSENTIAL QUESTION How do you combine, multiply, divide, and compose functions, and how do you find the domain of the resulting function?

SAMPLE STUDENT WORK

- A. $r(x) = 25x$
- B. $c(x) = 15x + 1000$
- C. No; grooming only 95 pets results in a loss of \$50 for Grooming USA. Grooming 110 pets results in a profit of \$100.
- D. To find the profit from 95 pets, I calculated $r(95) - c(95)$; to find the profit from 110 pets, I calculated $r(110) - c(110)$. If I needed to find the profit for several different values, it would make sense to find a function rule for profit and apply that simplified rule rather than applying two different functions and finding the difference of their values.



STEP 2 Understand & Apply

INTRODUCE THE ESSENTIAL QUESTION

Establish Mathematics Goals to Focus Learning **ETP**

Use operations with integers to discuss how combining integers by addition, subtraction, and multiplication always results in an integer, while combining integers by division does not always result in an integer. Tell students to compare those ideas to the results they see in this lesson when they combine functions by addition, subtraction, multiplication, and division.

EXAMPLE 1 Add and Subtract Functions

Pose Purposeful Questions **ETP**

- Q:** How do you combine functions by addition or subtraction?
[Combine the like terms of the two functions.]
- Q:** How do you know that the combined function has the same domain as the two original functions?
[The possible elements for the combined function are the same as the possible elements for the original functions.]
- Q:** What new notation is used in the example and what does it mean?
[($f + g$) x means $f(x) + g(x)$, and ($f - g$) x means $f(x) - g(x)$.]

Examples are available at SavvasRealize.com.

ESSENTIAL QUESTION
How do you combine, multiply, divide, and compose functions, and how do you find the domain of the resulting function?

EXAMPLE 1 Add and Subtract Functions

How do you define the sum, $f + g$, and the difference, $f - g$, of the functions $f(x) = 3x + 4$ and $g(x) = x^2 - 5x + 2$?

A. What is the sum of $f(x) = 3x + 4$ and $g(x) = x^2 - 5x + 2$?

To define the sum of two functions with known rules, add their rules.

$$(f + g)(x) = f(x) + g(x)$$

$$= (3x + 4) + (x^2 - 5x + 2)$$

$$= x^2 + (3x - 5x) + (4 + 2)$$

$$= x^2 - 2x + 6$$

The domain of f is $\{x \mid x \text{ is a real number}\}$.
The domain of g is $\{x \mid x \text{ is a real number}\}$.
So the domain of $f + g$ is $\{x \mid x \text{ is a real number}\}$.
The sum of the two functions is $x^2 - 2x + 6$.

CONTINUED ON THE NEXT PAGE

COMMUNICATE PRECISELY
Defining a function includes describing its domain. The domain of $f + g$ is the intersection of the domains of f and g .

LESSON 5.5 Function Operations 261

Additional Examples

Go back

ADDITIONAL EXAMPLE 4
ESSENTIAL QUESTION

Let $f(x) = 2x^2 - 3$ and let $g(x) = 4x - 2$. Find $f(g(x))$ and find $g(f(x))$.

$$f(g(x)) = 2(4x - 2)^2 - 3$$

$$= 2(16x^2 - 16x + 4) - 3$$

$$= 32x^2 - 32x + 5$$

$$g(f(x)) = 4(2x^2 - 3) - 2$$

$$= 8x^2 - 12 - 2$$

$$= 8x^2 - 14$$

SOLUTION

Example 4 Help students practice evaluating composite functions.

- Q:** What is the first step for each problem?
[Replace the expression inside the outer parentheses.]

Additional Examples are available at SavvasRealize.com.

Go back

ADDITIONAL EXAMPLE 6
ESSENTIAL QUESTION

A member currently has 75,000 awards points. In what order should the member take the bonuses?

First take 10,000 points, then take 5% bonus.

$$10,000 + 75,000 = 85,000$$

$$1.05(85,000) = 89,250$$

Take a 5% bonus, then add 10,000 points.

$$1.05(75,000) = 78,750$$

$$78,750 + 10,000 = 88,750$$

The overall bonus is 500 points more if you add the 10,000 points first, and then take the 5% bonus.

SOLUTION

Airline Awards Members!
Get Both Bonuses!
Add 10,000 award points!
Add 5% of your award points!

Example 6 Help students see how the order of two functions can affect the final value.

- Q:** How do you decide which bonus to take first? [The greater total will occur when 5% is taken on the additional 10,000 award points.]
- Q:** Why do the two orders have different outcomes?
[Adding 5% first yields less because you are adding 5% of a smaller number.]



Activity

EXAMPLE 1 CONTINUED

Try It! Answers

- $f + g = 2x^2 + 5x + 2$
- $f - g = 2x^2 + 9x - 4$

Elicit and Use Evidence of Student Thinking ETP

- Q:** A student said that $f + g$ and $f - g$ are shorthand for longer expressions. Explain what the student means.
 $[f + g$ is a shorter way to express $(2x^2 + 7x - 1) + (3 - 2x)$, and $f - g$ is a shorter way to express $(2x^2 + 7x - 1) - (3 - 2x).$]

EXAMPLE 2 Multiply Functions

Pose Purposeful Questions ETP

- Q:** How does the example involve multiplying functions?
 [The problem describes three functions—demand, price, and revenue—and revenue can be calculated by multiplying demand times price.]
- Q:** How do you multiply two functions?
 [Multiply each term of one function by each term of the other function and combine like terms]
- Q:** For the demand function $d(x)$ and the price function $p(x)$, is $d(x) \cdot p(x)$ equal to $p(x) \cdot d(x)$?
 [Yes; the products are equal because of the Commutative Property of Multiplication.]

Examples are available at SavvasRealize.com.

EXAMPLE 1 CONTINUED

B. What is the difference of $f(x) = 3x + 4$ and $g(x) = x^2 - 5x + 2$?

To define the difference of two functions with known rules, subtract their rules.

$$\begin{aligned}(f - g)(x) &= f(x) - g(x) \\ &= (3x + 4) - (x^2 - 5x + 2) \\ &= 3x + 4 - x^2 + 5x - 2 \\ &= -x^2 + 8x + 2\end{aligned}$$

The domain of f is $\{x \mid x \text{ is a real number}\}$. The domain of g is $\{x \mid x \text{ is a real number}\}$. So the domain of $f - g$ is $\{x \mid x \text{ is a real number}\}$.
 The difference of the two functions is $-x^2 + 8x + 2$.

Try It! 1. Let $f(x) = 2x^2 + 7x - 1$ and $g(x) = 3 - 2x$. Identify rules for the following functions.

a. $f + g$ b. $f - g$

EXAMPLE 2 Multiply Functions

The demand d , in units sold, for a company's new brand of cell phone at price x , in dollars, is $d(x) = 5,000 - 10x$. The cost to manufacture the cell phone is $c(x) = 2,200 + 80x$. What is the company's expected profit from cell phone sales in terms of the price, x ?

Profit Price Demand Cost

The company's profit will equal the price of its cell phones multiplied by the demand for its phones less the cost to manufacture.

Profit = price $\times$ demand - cost

The demand is the function $d(x)$. The price is the function $p(x)$.

Price:	$p(x) = x$	Domains	$p(x): 0 \leq x$
Demand:	$d(x) = 5,000 - 10x$	$d(x): 0 \leq x \leq 500$	
Cost:	$c(x) = 2,200 + 80x$	$c(x): 0 \leq x \leq 500$	
Profit:	$P(x) = p(x) \cdot d(x) - c(x)$	$P(x): 0 \leq x \leq 500$	

$P(x) = x(5,000 - 10x) - (2,200 + 80x)$
 $= 5,000x - 10x^2 - 2,200 - 80x$
 $= -10x^2 + 4,920x - 2,200$

Price, demand, and cost cannot be negative.

Domain is the intersection of the domains of p , d , and c .

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Extend Student Thinking

USE WITH EXAMPLE 3 Have students define and state the domains of the following quotients when $f(x) = x^2 + 2x - 3$, $g(x) = x^2 - 1$, and $h(x) = x^2 + 4x + 3$.

- $\frac{f}{g + h}$ $[\frac{x^2 + 2x - 3}{2x^2 + 4x + 2}]$. The domain is the set of all real numbers x such that $x \neq -1$.]
- $\frac{f - h}{g}$ $[\frac{-2x - 6}{x^2 - 1}]$. The domain is the set of all real numbers x such that $x \neq -1$ and $x \neq 1$.]
- $\frac{h - g}{g - f}$ $[\frac{2x + 2}{-x + 1}]$. The domain is the set of all real numbers x such that $x \neq 1$.]

- Q:** How do you find the zeros of each denominator? [Set the denominator equal to zero, and solve for the variable.]
- Q:** How do the zeros of the denominator affect the domain of the function? [The zeros of the denominator are excluded from the domain of the function.]

STEP 2 Understand & Apply



Try It! Answer

2. $d \cdot p = -0.25x^3 - 4x^2 + 1,000x + 16,000$

Elicit and Use Evidence of Student Thinking ETP

Q: Is it possible for x to have the value 64?

[No; if $x = 64$, then $d(64) = -0.25(64)^2 + 1,000 = -24$, and $d(x)$ cannot be less than 0.]

EXAMPLE 3 Divide Functions

Pose Purposeful Questions ETP

Q: How are the domains of g and $\frac{f}{g}$ different?

[g can have domain values that make the function equal zero, but those values cannot be in the domain of $\frac{f}{g}$.]

Try It! Answers

3. a. $\frac{f}{g} = x - 6$; The domain is all real numbers x such that $x \neq -3$.
 b. $\frac{f}{g} = \frac{1}{x+2}$; The domain is all real numbers x such that $x \neq 3$ and $x \neq -2$.

Elicit and Use Evidence of Student Thinking ETP

Q: Explain how you would find the domain of $\frac{g}{f}$.

[First, identify the domains of g and f . Then find the values that make $f(x)$ equal zero and eliminate them from the domain.]

Use with EXAMPLES 1–3

HABITS OF MIND

Communicate Precisely How are the domains of $f + g$, $f - g$, $f \cdot g$, and $\frac{f}{g}$ related to the domains of f and g ?

[If f and g have the same domain, the domains of $f + g$, $f - g$, and $f \cdot g$ are that same domain. The domain of $\frac{f}{g}$ is that same domain, but without any values for which g is zero. If f and g have different domains (as in Example 2), the domain for $f + g$, $f - g$, and $f \cdot g$ is the *intersection* of the two domains. The domain of $\frac{f}{g}$ is further restricted by removing any values for which g is zero.]

EXAMPLE 4 Compose Functions

Build Procedural Fluency From Conceptual Understanding ETP

Q: What is the difference between the two notations $f(x) \cdot g(x)$ and $f(g(x))$?

[$f(x) \cdot g(x)$ is the product of two expressions, and $f(g(x))$ uses the function $g(x)$ as the *variable* for the function $f(x)$.]



Examples are available at SavvasRealize.com.

EXAMPLE 2 CONTINUED

The profit the company will earn in terms of the cell phone price x is represented by $P(x) = -10x^2 + 4,920x - 2,200$.

Try It! 2. Suppose demand, d , for a company's product at cost, c , is predicted by the function $d(x) = -0.25x^2 + 1,000$, and price, p , that the company can charge for the product is given by $p(x) = x + 16$. Find the company's revenue function. Revenue is demand times price.

EXAMPLE 3 Divide Functions

How do you define the quotient $\frac{f}{g}$ of the functions $f(x) = x - 7$ and $g(x) = 2x^2 - 13x - 7$?

To define the quotient of two functions, take the quotient of their rules: $(\frac{f}{g})(x) = \frac{f(x)}{g(x)}$

$$\begin{aligned} (\frac{f}{g})(x) &= \frac{f(x)}{g(x)} = \frac{x-7}{2x^2-13x-7} \\ &= \frac{x-7}{(2x+1)(x-7)} \\ &= \frac{1}{2x+1} \end{aligned}$$

The quotient of $\frac{f}{g}$ is $\frac{1}{2x+1}$. The domain of $\frac{f}{g}$ is the set of all values for which f , g , and $\frac{f}{g}$ are defined, so g cannot be 0. This is the set of all real numbers x such that $x \neq 7$ and $x \neq -\frac{1}{2}$.

Try It! 3. Identify the rule and domain for $\frac{f}{g}$ for each pair of functions.

- a. $f(x) = x^2 - 3x - 18$, $g(x) = x + 3$
 b. $f(x) = x - 3$, $g(x) = x^2 - x - 6$

CONCEPTUAL UNDERSTANDING

LOOK FOR RELATIONSHIPS When finding the rule for $f(g(x))$, work from the inside parenthesis to the outside ones. Notice that $g(x)$ takes the place of the variable x in the function $f(x)$.

EXAMPLE 4 Compose Functions

Let $f(x) = x^2$ and let $g(x) = x + 1$. Investigate what happens when you apply the rule for g and then the rule for f to a number or variable.

A. Find the values of $f(g(0))$, $f(g(3))$, $f(g(-2))$.

For each value of x , first apply the rule for g . Then apply the rule for f to the result.

x	$g(x)$	$f(g(x))$
0	$g(0) = 0 + 1 = 1$	$f(g(0)) = 1^2 = 1$
3	$g(3) = 3 + 1 = 4$	$f(g(3)) = 4^2 = 16$
-2	$g(-2) = -2 + 1 = -1$	$f(g(-2)) = (-1)^2 = 1$

CONTINUED ON THE NEXT PAGE

LESSON 5-5 Function Operations 263



EXAMPLE 4 CONTINUED

Try It! Answers

4. a. $f(g(2)) = 11$
b. $f(g(x)) = 6x - 1$

Common Error

Try It! 4b When evaluating $f(3x)$, students may write $2(3x - 1)$ rather than $2(3x) - 1$. Encourage them to write the function as $f(\square) = 2(\square) - 1$ to emphasize that the expression that appears in the parentheses after the f should appear within parentheses on the right side of the equation in place of the variable x .

EXAMPLE 5 Write a Rule for a Composite Function

Pose Purposeful Questions ETP

- Q:** What notation is introduced in the example and what does it mean?
[The operation symbol $\circ$ in $f \circ g$ means “apply the rule for g , and then apply the rule for f .”]
- Q:** What procedure do you perform to find the rule for $f \circ g$?
[Substitute the expression $g(x)$ for the variable in $f(x)$, then apply the operations indicated by $f(x)$.]



Examples are available at SavvasRealize.com.

EXAMPLE 4 CONTINUED

B. Find the rule for $f(g(x))$.

$$\begin{aligned} g(x) &= x + 1 \\ f(g(x)) &= f(x + 1) \\ &= (x + 1)^2 \\ &= x^2 + 2x + 1 \\ f(g(x)) &= x^2 + 2x + 1 \end{aligned}$$

When you apply the rule for one function to the rule of another function, you create a new function.

Try It! 4. Let $f(x) = 2x - 1$ and $g(x) = 3x$. Find the value or identify the rule for the following functions.

- a. $f(g(2))$ b. $f(g(x))$

CONCEPT Composite Function

A **composite function** is the result of applying the rule for one function, f , to the rule of another function, g . The new rule is denoted as $f \circ g$.

$$(f \circ g)(x) = f(g(x))$$

The operation, $\circ$, that forms a composite function is called **composition of functions**.

The domain of $f \circ g$ is the set of all real numbers x in the domain of g such that $g(x)$ is in the domain of f .

So the domain of the composition is the intersection of the domains of g and $f \circ g$, but not f .

EXAMPLE 5 Write a Rule for a Composite Function

What is the rule for the composition $(f \circ g)(x)$?

- A. $f(x) = \sqrt{x + 7}$ and $g(x) = 2x - 5$

$$\begin{aligned} (f \circ g)(x) &= f(g(x)) \\ &= f(2x - 5) \\ &= \sqrt{(2x - 5) + 7} \\ &= \sqrt{2x + 2} \end{aligned}$$

The rule for the composition $(f \circ g)(x)$ is $\sqrt{2x + 2}$. The domain is $x \geq -1$, which is the intersection of the domains of g and $f \circ g$, which are all real numbers and $x \geq -1$.

CONTINUED ON THE NEXT PAGE



Support Struggling Students

USE WITH EXAMPLE 5 Help students focus on specific aspects of the example by discussing the following questions.

- Q:** What does the expression $f(g(x))$ in Part A mean?
[Evaluate $g(x)$ to get a number or an expression. Then evaluate the function f for that number or expression.]
- Q:** How is $g(f(x))$ in Part B different from $f(g(x))$?
[The difference is the order of using the functions. Evaluate the “inside” expression first, then use the result to evaluate the “outside” expression.]

STEP 2 Understand & Apply



Try It! Answers

5. a. $(f \circ g)(x) = x^3 + 3x^2 + 3x + 1$ and $(g \circ f)(x) = x^3 + 1$
 b. $(f \circ g)(x) = 3\sqrt{8x + 22}$ and $(g \circ f)(x) = 12\sqrt{2x} + 11$

Elicit and Use Evidence of Student Thinking ETP

Q: Part a: For the two given functions f and g , how can you tell when $f \circ g$ will be negative?

[$f(x) = x^3$, so $f(g(x))$ will be negative when $g(x)$ is negative. Because $g(x)$ is negative when $x + 1 < 0$, $f \circ g$ is negative when $x < -1$.]

EXAMPLE 6 Use a Composite Function Model

Pose Purposeful Questions ETP

Q: Why does order matter for discounts?

[The amount of a percent coupon depends on the base price. To maximize a percent coupon, take it off the larger price; that is, before a dollars-off coupon.]

Try It! Answer

6. Let x = purchase price.
 Discount: $d(x) = 0.8x$
 Tax: $t(x) = 1.06x$
 Actual price with discount: $t \circ d = 0.848x$

HABITS OF MIND

Use with EXAMPLES 4–6

Reason Let $f(x) = \frac{1}{3}(x - 2)$ and $g(x) = 3x + 2$. Find $f \circ g$ and $g \circ f$. Are they equivalent? If so, does this prove that the composition of two functions is commutative?

[$f(g(x)) = x$ and $g(f(x)) = x$, so in this case $f \circ g = g \circ f$. One example is not a proof, so this does not prove that composition of functions is commutative.]

Examples are available at SavvasRealize.com.

APPLICATION

STUDY TIP
Using a model can help solve a problem. By writing functions to model the discounts, determine which sequence of functions provides a better deal for the customer.

EXAMPLE 5 CONTINUED

B. $f(x) = x^2 + x + 2$ and $g(x) = 4 - x$

$$(f \circ g)(x) = f(g(x))$$

$$= f(4 - x)$$

$$= (4 - x)^2 + (4 - x) + 2$$

$$= 16 - 8x + x^2 + 4 - x + 2$$

$$= x^2 - 9x + 22$$

The rule for the composition $(f \circ g)(x)$ is $x^2 - 9x + 22$, and the domain is all real numbers.

Try It! 5. Identify the rules for $f \circ g$ and $g \circ f$.

a. $f(x) = x^3$, $g(x) = x + 1$

b. $f(x) = 3\sqrt{2x}$, $g(x) = 4x + 11$

EXAMPLE 6 Use a Composite Function Model

A clothing store posts discounts on social media. Followers are allowed to use multiple discounts. They simply need to tell the cashier in which order they want the discounts.

On their next trip to the store, in which order should they ask for the discounts to save the most money?

Write functions to model the discounts, letting x represent the price of the purchase.

$$f(x) = x - 5$$

$$g(x) = x - 0.1x$$

$$= 0.9x$$

price – 10% of price

10% off, then \$5 off $\Rightarrow$ Identify the rule for $f \circ g$.

$$(f \circ g)(x) = f(g(x))$$

$$= f(0.9x)$$

$$= 0.9x - 5$$

This is the better deal.

\$5 off, then 10% off $\Rightarrow$ Identify the rule for $g \circ f$.

$$(f \circ g)(x) = g(f(x))$$

$$= g(x - 5)$$

$$= 0.9(x - 5)$$

$$= 0.9x - 4.5$$

Suppose a shirt costs \$30.00 before the discounts. Applying the discounts as $(f \circ g)(x)$ means the new cost is $(0.9)(30) - 5 = 22$, or \$22.00. Applying the discounts as $(f \circ g)(x)$ means the new cost is $(0.9)(30) - 4.5 = 22.5$, or \$22.50. The first method yields the better deal.

Try It! 6. As a member of the Games Shop rewards program, you get a 20% discount on purchases. All sales are subject to a 6% sales tax. Write functions to model the discount and the sales tax, then identify the rule for the composition function that calculates the final price you would pay at Games Shop.

LESSON 5-5 Function Operations 265

ELL English Language Learners (Use with EXAMPLE 6)

LISTENING BEGINNING Point out the words *order* and *sequence* in the example and explain that they mean the same thing. Have students listen as you read the definition of *order*: “the arrangement of things following one after another, as in space or time.”

Q: How is *order* important in the problem? [The final price changes depending on which discount is used first.]

Q: What is an example of when you can *not* change the order of operations in a problem?

[When you simplify an expression, you must follow the same order of operations.]

SPEAKING DEVELOPING Choose some words from the example, such as *discounts*, *deal*, *customers*, *cashier*, etc. Have students take turns describing a word to the other students and have the others guess the word.

Q: How might a *customer* and a *cashier* interact?

[A cashier helps a customer buy items in a store; he rings up the purchases, and the customer gives the cashier money for the purchases.]

Q: How are *discounts* and *deals* related?

[A *discount* is money off the price of an item, and a *deal* is a good price on an item.]

WRITING EXPANDING There are some advanced verbs used in the example. Point out the words *applying*, *posts*, *identify*, and *yields*. Challenge students to write their own sentences correctly using these verbs.

Q: What does it mean to *post* on social media?

[to put a statement, advertisement, or photo online to share with other people]

Q: What is a word that means the same thing as the word *yields*, as it is used in this example?

[produces or gives]



CONCEPT SUMMARY Function Operations

Q: How do operations on functions relate to what you have learned before?

[Performing operations on functions is very similar to performing operations on integers, rational numbers, and polynomials.]

Do You **UNDERSTAND?** | Do You **KNOW HOW?**

Common Error

Exercise 10 Some students may believe that a zero of any function is not in the domain of a combination of functions. Encourage students to use a graphing utility to graph the combined function and look for asymptotes. The domain is only restricted for values where an asymptote occurs.

Answers

- To perform an arithmetic operation on a pair of functions, substitute their rules and perform the operation on them. The domain of the resulting function will be the set of all real numbers for which the original two functions *and* the resulting function are defined.
To compose functions, apply one function to the rule of the other. The domain of the resulting function will be the set of all real numbers for which both the composite function *and* the inner function (the first function applied) are defined.
- A composite function is a method of combining functions such that the output of one function is the input of the other function. Example: Given $f(x) = 2x$ and $g(x) = x + 1$, the composite function $f \circ g = f(g(x)) = 2(x + 1) = 2x + 2$.
- The domain is all real numbers except $x = -3$, for which $g(x)$ is equal to 0. $g(x)$ is in the denominator, so when $x = -3$, $\frac{f}{g}$ is undefined.
- The operations of subtraction and division are not commutative.
- $3x^2 + 7x$
- $3x^2 + 3x + 2$
- $-3x^2 - 3x - 2$
- $x^3 - 2x^2 - 7x - 4$
- $\frac{x^2 + 2x + 1}{x - 4}$ for all real numbers x , such that $x \neq 4$
- $\frac{x - 4}{x^2 + 2x + 1}$ or $\frac{x - 4}{(x + 1)^2}$ for all real numbers x , when $x \neq -1$
- $18x^2 + 5$



Concept Summary, Do You Understand, and Do You Know How are available at SavvasRealize.com.

CONCEPT SUMMARY Function Operations

	Add or Subtract Functions	Multiply or Divide Functions	Compose Functions
ALGEBRA	$(f + g)(x) = f(x) + g(x)$ $(f - g)(x) = f(x) - g(x)$	$(f \cdot g)(x) = f(x) \cdot g(x)$ $\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$	$(f \circ g)(x) = f(g(x))$ $(g \circ f)(x) = g(f(x))$
WORDS	The domain of the sum or difference of f and g is the intersection of the domain of f and the domain of g .	The domain is the set of all real numbers for which f and g and the new function are defined.	The domain of $f \circ g$ is the set of all real numbers x , in the domain of g , such that $g(x)$ is in the domain of f .
NUMBERS	Let $f(x) = 3x + 5$ and $g(x) = x - 3$ $f + g = (3x + 5) + (x - 3)$ $= 4x + 2$ $f - g = (3x + 5) - (x - 3)$ $= 2x + 8$	Let $f(x) = 3x + 5$ and $g(x) = x - 3$ $f \cdot g = (3x + 5)(x - 3)$ $= 3x^2 - 4x - 15$ $\frac{f}{g} = \frac{3x + 5}{x - 3}$ for $x \neq 3$	Let $f(x) = 3x + 5$ and $g(x) = x - 3$ $f \circ g = 3(x - 3) + 5$ $= 3x - 4$ $g \circ f = (3x + 5) - 3$ $= 3x + 2$

Do You UNDERSTAND?

- ESSENTIAL QUESTION** How do you combine, multiply, divide, and compose functions, and how do you find the domain of the resulting function?
- Vocabulary** In your own words, define and provide an example of a composite function.
- Error Analysis** Robie said the domain of $\frac{f}{g}$ when $f(x) = 5x^2$ and $g(x) = x + 3$ is the set of real numbers. Explain why Robie is incorrect.
- Use Structure** Explain why changing the order in which two functions occur affects the result when subtracting and dividing the functions.

Do You KNOW HOW?

- Let $f(x) = 3x^2 + 5x + 1$ and $g(x) = 2x - 1$.
- Identify the rule for $f + g$.
 - Identify the rule for $f - g$.
 - Identify the rule for $g - f$.
 - Let $f(x) = x^2 + 2x + 1$ and $g(x) = x - 4$. Identify the rule for $f \cdot g$.
 - Identify the rule for $\frac{f}{g}$, and state the domain.
 - Identify the rule for $\frac{g}{f}$, and state the domain.
 - If $f(x) = 2x^2 + 5$ and $g(x) = -3x$, what is $f(g(x))$?

STEP 3 Practice & Problem Solving



PRACTICE & PROBLEM SOLVING

Lesson Practice You may opt to have students complete the automatically scored Practice and Problem Solving items online powered by MathXL for School.

Choose from: Lesson Practice

Adaptive Practice

You may also take advantage of the bank of exercises for assigning additional practice.

Assignment Guide

On-Level	Advanced
13, 19–23, 33	12–36

Engage Through Student Choice

Promote student agency by allowing students to choose practice items. You may structure this choice in many ways. For example:

Assign each section a point value. Students choose at least one item from each section and items chosen should have a minimum of 20 total points.
Understand, Apply 2 points each
Practice 1 point each
Assessment Practice 1 point each
Performance Task 3 points

Item Analysis

Example	Items	DOK
1	20–21	1
	19, 33	2
2	22	2
	23	1
3	13	2
	24–27, 35	1
4	18, 31, 36	2
	28, 29	1
5	12, 14, 34	2
	15, 17, 30	2
6	16, 32	3



Practice & Problem Solving and video tutorials are available at SavvasRealize.com.

Scan the QR code to access a video tutorial.

PRACTICE & PROBLEM SOLVING

UNDERSTAND

- Generalize** Does $f \circ g$ always equal $g \circ f$? Justify your response.
- Construct Arguments** Explain why the domain for the quotient of functions might not be the set of all real numbers.
- Error Analysis** Describe and correct the error a student made in finding the rule for the composition $f \circ g$ of the functions $f(x) = 3x^2 - x + 2$ and $g(x) = 2x + 1$.

$$\begin{aligned}
 f \circ g &= f(g(x)) \\
 &= 3(2x + 1)^2 - 2x + 1 + 2 \\
 &= 3(4x^2 + 4x + 1) - 2x + 1 + 2 \\
 &= 12x^2 + 12x + 3 - 2x + 1 + 2 \\
 &= 12x^2 + 10x + 6
 \end{aligned}$$



- Make Sense and Persevere** Identify the rules for two functions, $f(x)$ and $g(x)$, for which $f \circ g = g \circ f$.
- Higher Order Thinking** Suppose two functions, $f(x)$ and $g(x)$ are only defined by the ordered pairs listed below.
 $f = \{(6, 7), (5, 2), (4, 1), (10, 8)\}$
 $g = \{(5, 4), (3, 6), (1, 5), (2, 10)\}$
 Find the ordered pairs that comprise $(f \circ g)(x)$.
 $(f \circ g)(x) = \{(5, 1), (3, 7), (1, 2), (2, 8)\}$
- Mathematical Connections** Relate evaluating $(f \circ g)(3)$ to finding the composition rule $(f \circ g)(x)$. What are the benefits of each?
- Make Sense and Persevere** Identify rules for two functions $f(x)$ and $g(x)$, for which $f(g(x)) = x$.
- Construct Arguments** Is it possible that $f(x) - g(x) = c$, where c is a constant? If so, give an example. What must be true about two linear functions? If not, explain why it is not possible.
 It is possible; Sample: If the graphs of the two linear functions are parallel, then their difference will be a horizontal line; $f(x) = 3x - 4$ and $g(x) = 3x + 2$, so $f(x) - g(x) = -6$, which represents a horizontal line.

PRACTICE

Let $f(x) = \frac{5}{3}x^2 + 2x - \frac{5}{8}$ and $g(x) = 3x^2$. Identify the rules for the following functions. SEE EXAMPLE 1

$$\begin{aligned}
 20. f + g & \quad 21. f - g \\
 f + g &= \frac{14}{3}x^2 + 2x - \frac{5}{8} \quad f - g = -\frac{4}{3}x^2 + 2x - \frac{5}{8}
 \end{aligned}$$

- Suppose the demand d , in units sold, for a company's jeans at price x , in dollars, is $d(x) = 6,500 - 6.83x$.

- If revenue = price $\times$ demand, write the rule for the function $R(x)$, which represent the company's expected revenue in jeans sales. Then state the domain of this function.
 $R(x) = 6,500x - 6.83x^2$ for $0 \leq x \leq 951.68$
- If the cost to manufacture the jeans is $c(x) = 386 + 1.27x$, find the equation for the company's profit. How much does the company earn if the price is \$79?

SEE EXAMPLE 2

- Identify the rule and domain for $\frac{f}{g}$ when $f(x) = 10x^2 + 3x - 18$ and $g(x) = 2x + 3$.

SEE EXAMPLE 3

Let $f(x) = 4x - 5$ and $g(x) = -7x$. Evaluate each expression. SEE EXAMPLE 4

$$\begin{aligned}
 24. f(g(3)) &= -89 & 25. f(g(x)) &= -28x - 5 \\
 26. g(f(2)) &= -21 & 27. g(f(x)) &= -28x + 35
 \end{aligned}$$

Let $f(x) = x^2 + x$ and $g(x) = 9 - 2x$. Identify the rules for the following functions. SEE EXAMPLE 5

$$28. f \circ g \quad 4x^2 - 38x + 90 \quad 29. g \circ f \quad -2x^2 - 2x + 9$$

- A sporting goods store is running a summer sale on its snowboards. Kadia is interested in a snowboard that normally costs \$400. The store is offering two special offers.

In which order should these special offers be applied to the cost of the snowboard in order to benefit Kadia? Explain. SEE EXAMPLE 6



LESSON 5-5 Function Operations 267

Answers

- In general $f \circ g$ and $g \circ f$ are usually not equal. Sample: If $f(x) = x + 1$ and $g(x) = 4x$, then $(f \circ g)(x) = 4x + 1$ and $(g \circ f)(x) = 4x + 4$.
- because there are restrictions on the x -values in the denominator of a fraction such that it cannot be defined
- When $2x + 1$ was substituted into $3x^2 - x + 2$ for x and when $-x$ was replaced with $2x + 1$, the negative sign was not distributed.
- Answers may vary. Sample: When $f(x) = 2x$ and $g(x) = 5x$, $(f \circ g)(x) = 10x$ and $(g \circ f)(x) = 10x$.
- $(f \circ g)(x) = \{(5, 1), (3, 7), (1, 2), (2, 8)\}$

See answers for 17–18, 22–23, 30 on the next page.



Practice

Answers

17. Evaluating function composition for a specific value is finding $g(3)$ and then substituting $g(3)$ for x in $f(x)$. This approach is good when you only want to find a few values. Finding the rule of $(f \circ g)(x)$ is similar in that you evaluate $f(x)$ for the rule $g(x)$, that is $f(g(x))$. You can then recognize key features of the new function, including domain, range, and end behavior.
18. Sample answer: If $f(x) = \frac{1}{2}x$ and $g(x) = 2x$, then $f(g(x)) = x$.
22. b. $P(x) = -6.83x^2 + 6,498.73x - 386$; $P(79) = \$470,387.64$
23. $\left(\frac{f}{g}\right)(x) = 5x - 6$ for all real numbers x , such that $x \neq -\frac{3}{2}$
30. It is better for Kadia if the 10% discount is applied before the \$50 instant rebate because he will save \$5 more.
31. a. $C(h(x)) = 600x + 500$
b. \$5,300
c. $C(h(x))$ represents the cost to produce x hours' worth of shovels.
32. a. \$72.25
b. \$71.50
c. Since $\$71.50 < \72.25 , it is a better deal for the consumer if the 15% discount is applied prior to the \$5 off discount.
33. a. subtraction
b. $p(x) = b(x) - d(x) = -15x + 295$; the domain is the set of all real numbers x such that $0 \leq x \leq 15$.
36. Part A $K = \frac{5}{9}(F - 32) + 273$
Part B 270 K



Practice & Problem Solving is available at SavvasRealize.com.

PRACTICE & PROBLEM SOLVING

APPLY

31. **Model With Mathematics** The cost (in dollars) to produce x shovels in a factory is given by the function $C(x) = 20x + 500$. The factory produces 30 shovels in one hour.
a. Find the rule for $C(h(x))$, where the function $h(x)$ is the number of shovels produced in x hours.
b. Find the cost when $h = 8$ hours.
c. Explain what the function $C(h(x))$ represents.
32. **Use Structure** A music store is running the following promotions.

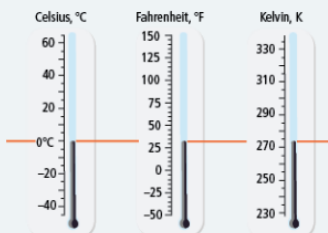


- a. Use composition of functions to find the sale price of a \$90 purchase when the \$5 off discount is applied prior to the 15% off discount.
- b. Use composition of functions to find the sale price of a \$90 purchase when the 15% off discount is applied prior to the \$5 off discount.
- c. In which order is the deal better for the consumer? Explain.
33. **Reason** From 2000 to 2015, the number of births, b , (in the hundreds) in Fairfield County can be modeled by the function $b(x) = 300 - 5x$. The number of deaths, d , (in the hundreds) can be modeled by the function $d(x) = 10x + 5$. The variable x represents the number of years since 2000.
a. Which function operation can be used to represent the net increase in the population?
b. Write and simplify a function which represents the net increase in the population, p , against x , the number of years since 2000. State the domain of this function.

ASSESSMENT PRACTICE

34. Given that $f(x) = x^2 + 8x + 3$ and $g(x) = -x - 7$, which of the following are true? Select all that apply.
☒ $f + g = x^2 + 7x - 4$
☒ $f(g(x)) = x^2 + 6x - 4$
☐ The domain of $\frac{f}{g}$ is the set of all real numbers.
☐ $f(x) \cdot g(x) = -x^3 - 15x^2 + 53x + 21$
☒ In the composition $g \circ f$, the output $f(x)$ is used as the input for g .
35. **SAT/ACT** Find the value of $f(g(5))$ if $f(x) = 4x + 1$ and $g(x) = x^2 + 6$.
Ⓐ 101 Ⓑ 124 Ⓒ 125 Ⓓ 676 Ⓔ 682

36. **Performance Task** The temperature in degrees Celsius is 32 less than the Fahrenheit temperature, multiplied by five-ninths. The temperature in degrees Kelvin is the number of degrees Celsius plus 273.



Part A Derive a conversion formula for finding the number of degrees Kelvin, given the temperature in Fahrenheit.

Part B Using your conversion formula from part (a), find the temperature in degrees Kelvin when the temperature is 27°F. Round to the nearest whole number if necessary.



LESSON QUIZ

Use the Lesson Quiz to assess students' understanding of the mathematics in the lesson.

Students can take the Lesson Quiz online or you can download a printable copy from [SavvasRealize.com](https://www.savvasrealize.com). The Lesson Quiz is also available in the *Assessment Sourcebook*.

Item Analysis

Item	DOK	Skills Review and Practice
1	3	F52, F26
2	3	F53, F26
3A	3	F53, F26
3B	2	F53, F26
4	3	F53, F26
5	2	F64, F26



Use the student scores on the Lesson Quiz to prescribe differentiated assignments.

If students take the Lesson Quiz online, it will be automatically scored and appropriate differentiated practice will be assigned based on student performance.

I Intervention	0–3 points	- Reteach to Build Understanding - Mathematical Literacy and Vocabulary - Lesson Virtual Nerd videos
O On-Level	4 points	Enrichment
A Advanced	5 points	Enrichment



Lesson Quiz is available at [SavvasRealize.com](https://www.savvasrealize.com).

ASSESSMENT RESOURCES

Name _____

enVision Algebra 2

SavvasRealize.com

5-5 Lesson Quiz

Function Operations

1. Let $f(x) = 3x^2 - 2x + 6$ and $g(x) = 7x - 4$. Identify the rule for $g(f(x)) + g(x)$.

- ☒ A $21x^2 - 7x + 34$
☐ B $21x^2 - 21x + 50$
☐ C $147x^2 - 182x + 62$
☐ D $21x^3 - 26x^2 + 57x - 28$

2. The demand d for a company's product at cost x is predicted by the function $d(x) = 500 - 2x$. The price p in dollars that the company can charge for the product is given by $p(x) = x + 5$. Use the formula Revenue = price $\times$ demand to find the revenue function for the product.

$$R(x) = \boxed{2,500} + \boxed{490}x - \boxed{2}x^2$$

Part A

Identify the rule for $\frac{f}{g}$ when $f(x) = -3x - 6$ and $g(x) = x^2 - x - 6$.

- ☐ A $\frac{f(x)}{g(x)} = \frac{-2}{(x+2)}$ ☐ C $\frac{f(x)}{g(x)} = \frac{3}{(x+2)}$
☐ B $\frac{f(x)}{g(x)} = \frac{-3(x-2)}{(x-6)(x+1)}$ ☒ D $\frac{f(x)}{g(x)} = \frac{-3}{x-3}$

Part B

Select all values that are NOT in the domain of $\frac{f}{g}$.

- ☐ A. -1 ☐ D. 6
☒ B. 3 ☒ E. -2
☐ C. -3

4. Let $h(x) = \sqrt{x+3}$ and $k(x) = 2x+7$. Find the value $\frac{h(k(3)) - k(h(1))}{k(h(-2)) - h(k(-3))}$.

- ☐ A $\frac{2\sqrt{6} - 2\sqrt{3}}{\sqrt{6} - 7}$ ☐ C $2\sqrt{6} - \sqrt{12} + 7$
☐ B 4 ☒ D -1

5. Let $f(x) = 3x^2 + 2$ and $g(x) = \sqrt{x-4}$. What is the rule for the composition $f \circ g$?

- ☐ A $(f \circ g)(x) = \sqrt{3x^2 - 2}$, domain is all real numbers
☐ B $(f \circ g)(x) = \sqrt{3x^2 - 2}$, domain is $x \geq 4$
☐ C $(f \circ g)(x) = 3x - 10$, domain is all real numbers
☒ D $(f \circ g)(x) = 3x - 10$, domain is $x \geq 4$

enVision® Algebra 2 • Assessment Sourcebook



DIFFERENTIATED RESOURCES

I = Intervention **O** = On-Level **A** = Advanced

Reteach to Build Understanding **I**

Provides scaffolded reteaching for the key lesson concepts.

MathXL® for School

5-5 Reteach to Build Understanding

Function Operations

1. Complete the table to practice each operation.

	Step 1: Rule	Step 2: Equations	Step 3: Set Up	Step 4: Answer
Add	$f(x) + g(x)$	$f(x) = x + 1$ $g(x) = 2x + 11$	$x + 1 + 2x + 11$	$= 3x + 12$
Subtract	$f(x) - g(x)$	$f(x) = 4x + 8$ $g(x) = 2x + 5$	$4x + 8 - (2x + 5)$ $4x + 8 - 2x - 5$	$= 2x + 3$
Multiply	$f(x) \cdot g(x)$	$f(x) = x + 3$ $g(x) = 2x + 2$	$(x + 3)(2x + 2)$ $2x^2 + 2x + 6x + 6$	$= 2x^2 + 8x + 6$
Divide	$\frac{f(x)}{g(x)}$	$f(x) = x^2 + 4x + 4$ $g(x) = x + 2$	$\frac{x^2 + 4x + 4}{x + 2}$	$= x + 2$
Compose	$f(g(x))$	$f(x) = 2x + 3$ $g(x) = 3x + 2$	$2(3x + 2) + 3$ $6x + 4 + 3$	$= 6x + 7$

2. Fatima was asked to write the rule for $f \circ g$. She set it up as $(f \circ g)(x)$. What error did she make? Choose the correct answer: **a**

a. She set up the rule for multiplying the two equations.

b. She set up the rule for adding the two equations.

c. She set up the rule for dividing the two equations.

d. She set up the rule for subtracting the two equations.

How should she have set it up? Choose the correct answer: **c**

a. When composing them, she should have set them up as $\frac{f(x)}{g(x)}$.

b. When composing them, she should have set them up as $f(g(x))$.

c. When composing them, she should have set them up as $f(g(x))$.

d. When composing them, she should have set them up as $f(x) \cdot g(x)$.

3. What is the rule for the composition $f \circ g$?

$f(x) = 2x + 3$ and $g(x) = 3x + 2$

$f \circ g = f(g(x))$

$= 2(3x + 2) + 3$

$= 6x + 4 + 3$

$= 6x + 7$ Domain: $x \geq -\frac{1}{2}$

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Additional Practice **I O**

Provides extra practice for each lesson.

MathXL® for School

5-5 Additional Practice

Function Operations

Let $f(x) = 3x^2 - 5x - 11$ and $g(x) = 7 - 4x$. Identify rules for the following functions.

1. $f + g$

$f(x) + g(x) = 3x^2 - 12x - 4$

2. $f - g$

$f(x) - g(x) = 3x^2 - 5x - 18$

3. Suppose demand d for a company's product at cost x is predicted by the function $d(x) = 0.36x^2 + 810$, and that the price p that the company can charge for the product is given by $p(x) = x + 14$. Find the company's revenue function.

$R(x) = 0.36x^3 + x + 796$

4. Identify the rule and domain for f when $f(x) = x^2 - 5x - 36$ and $g(x) = x - 9$.

$(\frac{f}{g})(x) = x + 6$, all real numbers excluding 9

Let $f(x) = 3x - 2$ and $g(x) = 5x$. Identify the rule for the following functions.

5. $f(g(x))$

$15x - 2$

6. $f(g(x))$

$15x - 2$

7. Identify the rules for $f + g$ and $g \circ f$ when $f(x) = 2x^2 + 3$ and $g(x) = x - 1$.

$(f + g)(x) = 2x^2 + 2x + 2$

$(g \circ f)(x) = 2(x - 1)^2 + 3 = 2x^2 - 4x + 5$

8. As a member of the Game Shop rewards program, you get a 12% discount on purchases. All sales are subject to an 8% sales tax. Write functions to model the discount and the sales tax, then identify the rule for the composition function that calculates the final price you pay Games Shop.

$D(x) = x - 0.12x$; $T(x) = 0.08(x - 0.12x)$; $P = 0.9504x$

9. Describe and correct the error a student made in finding the rule for the composition $f \circ g$ when $f(x) = 2x^2 - 3x + 1$ and $g(x) = 2x - 1$.

The student did not replace the x with $(2x - 1)$.

$(f \circ g)(x) = f(g(x))$

$= 2(2x - 1)^2 - 3(2x - 1) + 1$

$= 2(4x^2 - 4x + 1) - 6x + 3 + 1$

$= 8x^2 - 14x + 6$

The cost in dollars to produce x shovels in a factory is given by the function $C(x) = 23x + 480$. The number of shovels that can be produced in h hours is given by the function $N(h) = 30h$.

10. Find the rule for $C(N(h))$.

$690h + 480$

11. Find the cost when $h = 9$ hours.

6600

Let $f(x) = 3x^2 + 2x - 3$ and $g(x) = 2x + 4$. Identify the rules for the following functions.

12. $f + g$

$3x^2 + 4x + 1$

13. $f - g$

$3x^2 - 7$

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Enrichment **O A**

Presents engaging problems and activities that extend the lesson concepts.

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5-5 Enrichment

Function Operations

Use compositions of the following functions to fill cells 1-6.

$f(x) = x^2 - 1$

$g(x) = x + 1$

$h(x) = x - 1$

$k(x) = x^2 + 1$

Rules of the Game:

1. You can fill each cell only with one composition of the functions.

2. You can use a function in more than one composition.

3. You cannot use the same composition more than once.

4. The sum of terms in each column and row with more than one cell is equal to $3x^2$.

5. Function $f(x) = x^2 - 1$ must be used in all compositions of the left column.

1	$f \circ g$				
2	$f \circ h$				
3	$g \circ f$	4	$g \circ h$	5	$h \circ f$
				6	$h \circ g$
				7	$h^2 - 2x$

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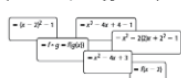
Mathematical Literacy and Vocabulary **I O**

Helps students develop and reinforce understanding of key terms and concepts.

5-5 Mathematical Literacy and Vocabulary

Function Operations

Ashton wanted to find a function rule for $f \circ g$, given $f(x) = x^2 - 1$ and $g(x) = x - 2$. He wrote these steps on note cards, but they got mixed up.



Organize the cards to complete the steps Ashton used to find the function rule.

1. First, $f \circ g = f(g(x))$

2. Second, $= f(x - 2)$

3. Next, $= (x - 2)^2 - 1$

4. Next, $= x^2 - 4x + 4 - 1$

5. Then, $= x^2 - 4x + 3$

6. Finally, $= x^2 - 4x + 3$

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Digital Differentiated Resources

The Reteach to Build Understanding, Additional Practice, and Enrichment activities are available as digital assignments. These activities are automatically assigned when students complete the lesson quiz online and are automatically scored.

